

Multiplication Operator

$$K(t): L^2 \rightarrow L^2$$

- Infinite-dimensional eigenspaces
- Pure essential spectrum
 - Not Fredholm
 - SF undefined

Base Elliptic Operator

$\mathcal{L}_0$ (self-adjoint, Fredholm)

Perturbed Elliptic Operator

$$\mathcal{A}(t) = \mathcal{L}_0 + K(t)$$

- Discrete spectrum near 0
- Finite-dimensional eigenspaces
 - Fredholm
 - SF well-defined